

Customer Segmentation for Personalized Promotion in E-Commerce

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Team Introduction



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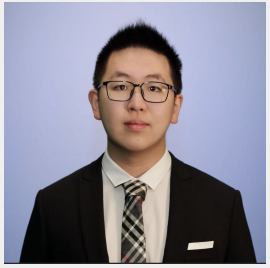
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Business Analyst



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Data Engineer



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Data Scientist



Agenda

- Business Context
- Population Characteristics
- Promotion Strategies
- Business Value
- Conclusion

Why is Personalization Important?

Individuals behave differently

Different customers have different characteristics and preferences

➔ *One promotion cannot fit everyone*

Drive Engagement

Personalized marketing drives stronger engagement and higher conversion.

➔ *Higher add-to-cart and purchase rates*

Proven impact in e-commerce

Amazon: **35% of sales** come from personalized recommendations.

➔ *Creates measurable revenue*



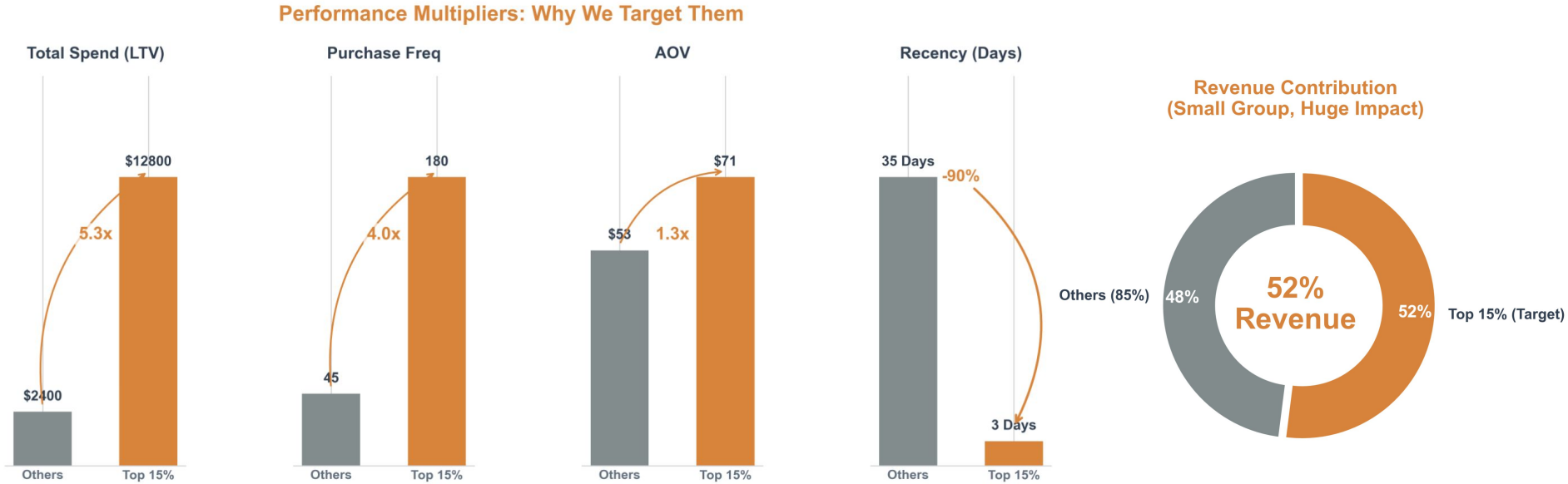
Our Proposal:

- After meeting with our CTO:
  - Clustering identifies the **top 15%** most engaged customers as the high-impact segment.
  - Targeted promotions to this group can drive **~20%** incremental sales.
  - Other segments still show a reliable **3–5% lift**, validating the strategy's broad effectiveness.



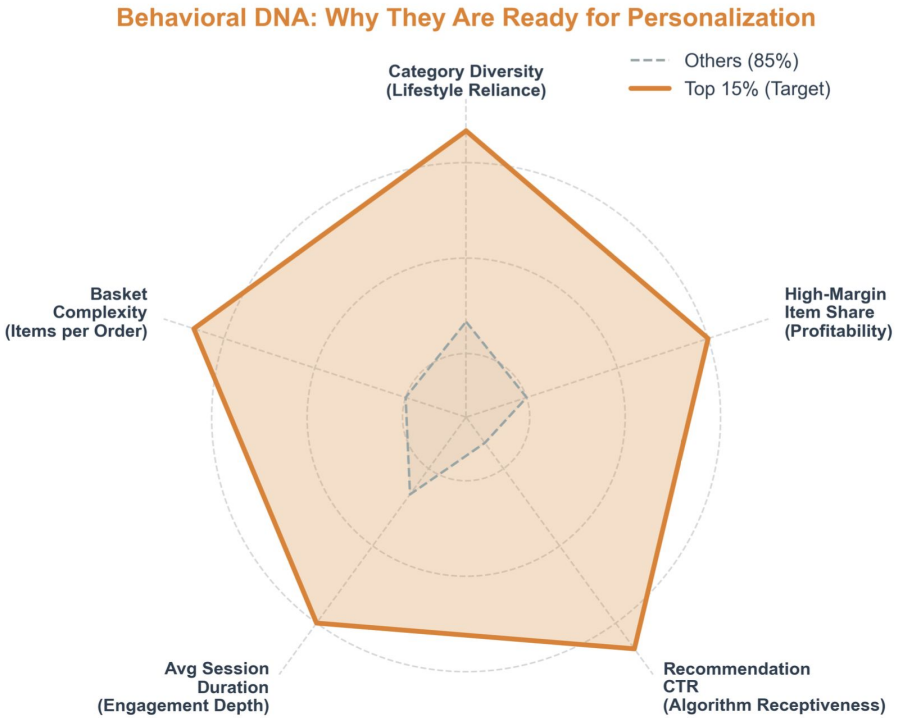
Descriptions of top 15% characteristics

While comprising only 15% of the population, this segment contributes **52% of total revenue**. Their value is driven by frequency (**4.0x engagement**) rather than just price, creating a consistent, high-fidelity data stream (**Recency: 3 days**) that significantly de-risks our personalization strategy.



Descriptions of top 15% characteristics

- **Category Diversity:** They show high **Lifestyle Reliance** by shopping across a wide range of categories.
- **High-Margin Item Share:** They drive superior **Profitability** by purchasing premium products.
- **Recommendation CTR:** They exhibit strong **Algorithm Receptiveness**, responding actively to suggestions.
- **Avg Session Duration:** They demonstrate deep **Engagement Depth** with significantly longer visits.
- **Basket Complexity:** They consistently place larger, multi-item orders.



Customer segmentation overview

	Behavioral Profile	Shopping Pattern	Strategic Focus
 **VIP Champions** (Top 15%)	• 4.0x frequency • 3-day recency • Premium buyers	• Cross-category buyers • High basket complexity • Algorithm-responsive	Hyper-personalization
 **Emerging Loyalists** (~25%)	• Medium-high engagement • Increasing frequency • Growing AOV	• Category-focused • Building brand loyalty	VIP elevation
 **Occasional Shoppers** (~35%)	• Seasonal purchases • 30+ day cycles • Price-sensitive	• Deal-motivated • Single-category focus	Activation
 **At-Risk** (~25%)	• Declining engagement • 60+ day recency	• Infrequent visits • Unpredictable behavior	Re-engagement

Source: PLAYINGPANDAS Analysis

Targeted promotion strategies



VIP Champions
(Top 15%)

Premium experience and exclusive benefits

- 48-hour early access to new and limited products
- Enhanced seasonal promotions (VIP only)
- 3x loyalty points on all purchases
- Premium free shipping (lower threshold)
- Priority customer service access



Emerging Loyalists
(~25%)

Engagement and loyalty acceleration

- 24-hour early access (one tier below VIP)
- 2× loyalty points during promotions
- “Unlock VIP status” incentives



Occasional Shoppers
(~35%)

Activation and frequency building

- Cart abandonment recovery emails
- %-off incentive on next purchase
- Free shipping above standard threshold



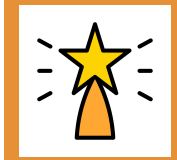
At-Risk
(~25%)

Win-back and re-engagement

- “We miss you” personalized email series
- Comeback offer (20% off next order)

Source: PLAYINGPANDAS Analysis

Business Value

	**VIP Champions** (Top 15%)	***Premium experience and exclusive benefits***	• 48-hour early access to new and limited products • Enhanced seasonal promotions (VIP only) • 3x loyalty points on all purchases • Premium free shipping (lower threshold) • Priority customer service access
	**Emerging Loyalists** (~25%)	***Engagement and loyalty acceleration***	• 24-hour early access (one tier below VIP) • 2× loyalty points during promotions • “Unlock VIP status” incentives
	**Occasional Shoppers** (~35%)	***Activation and frequency building***	• Cart abandonment recovery emails • %-off incentive on next purchase • Free shipping above standard threshold
	**At-Risk** (~25%)	***Win-back and re-engagement***	• “We miss you” personalized email series • Comeback offer (20% off next order)

Source: PLAYINGPANDAS Analysis

Business Value

Conclusion



Appendices

Appendix 1: xxxx

Appendix 2: xxx

Appendix 3: xxx